MIL-B-60124(MU)
AMENDMENT 2
12 September 1968
SUPERSEDING
AMENDMENT 1
30 April 1968

MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635412
(CANT UNIT)

This amendment forms a part of Military Specification MIL-B-60124(MU), dated 5 June 1964.

Paragraph 2.1, SPECIFICATIONS: Delete "MIL-I-45208-Inspection System Requirements".

Paragraph 3.4.1: Delete last sentence.

Add new paragraph 3.4.4 as follows:

"3.4.4 Storage temperature.- The cant unit shall show no evidence of physical failure and shall meet all requirements of this specification at a standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F."

Paragraph 3.5.2: Delete "plus twice the rated voltage".

Paragraph 3.6.2: Delete and substitute

"3.6.2 Magnet arm damping.- The magnet arm shall return to its normal gravity controlled position within a maximum of one cycle of oscillation when displaced at least 5 degrees."

Paragraph 3.6.3: Delete last sentence and substitute

"The minimum null indication, neglecting test circuit switching spikes, shall not exceed 8.0 millivolts peak-to-peak for cosine readings and 2.0 millivolts peak-to-peak for sine readings."

Paragraph 4.4.1: Delete and substitute

"4.4.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109."

FSC 1220

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Add new paragraphs 4.1.2, 4.1.2.1, and 4.1.2.2 as follows:

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) cant units plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

Paragraph 4.2.2, Table I: Add the following characteristics:

"106.	Dielectric strength	3.5.2	4.5.3
107.	Insulation resistance	3.5.3	4.5.4"

Paragraph 4.3: Delete "Control tests and substitute "Special sampling".

Paragraph 4.3.1, last sentence: Delete "control"; In tabulation heading delete "Control test" and substitute "Characteristic".

Paragraph 4.3.2: Delete second sentence and substitute "Each sample shall meet the following requirements and tests, and shall then meet the requirements and tests in Table I."; Delete tabulation and substitute

"Characteristic		Requirement	Test Procedure
304.	Storage temperature (-80° and +160°F)	3.4.4	4.5.1.4
305.	Resolver input travel	3.4.1, 3.6.1	4.5.1.1, 4.5.5.1
306.	Magnet arm damping	3.4.1, 3.6.2	4.5.1.1, 4.5.5.2
307.	Output voltages	3.4.1, 3.6.3	4.5.1.1, 4.5.5.3
308.	Water resistance (+60° to +90°F)	3.4.3	4.5.1.3

NOTE: The tests for Characteristics 305 through 307 inclusive shall be conducted at -40° and +150°F.

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Paragraph 4.3.3: Delete and substitute

"4.3.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirement, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved (see 4.2.3)."

Paragraph 4.4, first sentence: Delete "Measuring and Test Equipment of MIL-I-45208 and all".

* Paragraph 4.4.2: In Item No. 2, Description: Delete last sentence and substitute "Input impedence - 1 megohm minimum"; In Item No. 4, Description: Delete and substitute "Shall mount the cant unit in a true horizontal position. Rotate the cant unit up to 20° in either direction about the resolver axis with settings to 5°, 10° and 15° within 20 seconds. The test fixture shall also contain a setting of 15 degrees 3 minutes. Provisions for instantaneous release from a 20° position and return to the horizontal position without overtravel."

Paragraph 4.5.1.2: Delete last sentence and substitute

"The cant unit shall then be subjected to the test specified in Table I to determine conformance with the applicable requirements referenced therein."

Add new paragraph 4.5.1.4 as follows:

"4.5.1.4 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.4. The cant unit shall be exposed to each temperature extreme specified in 3.4.4 for a minimum of 5 hours. After each temperature exposure the cant unit shall be stabilized in an ambient temperature between 60° and 90°F and subjected to a visual, tactile and audible examination for evidence of physical failure. The cant unit shall then be subjected to the tests in 4.5.1.1 through 4.5.5.3 inclusive to determine conformance to 3.4.4. Caution shall be exercised during the temperature cycling to avoid subjecting the cant unit to thermal shock."

* Paragraph 4.5.5.2: Delete and substitute

"4.5.5.2 Magnet arm damping.- To determine conformance to 3.6.2, physically rotate the cant housing, in either direction, to a point at least 20 degrees about the centerline of the resolver axis by means of the test fixture. At this point of rotation, instantaneously release the test fixture to permit it to return to zero cant, without overtravel. The magnet arm shall return to its normal gravity controlled position within one cycle of oscillation as specified in 3.6.2. One cycle of

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oscillation shall be determined by observing the oscilloscope null on the test circuit."

*The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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